.1.1 the spaces are equipped with gas detection systems which activate the alarm at not more than 20% LFL and shut down the isolation valves, as appropriate, at not more than 40% LFL (see sections 16.4.2 and 16.4.8 of the Code); and

.1.2 the spaces are adequately ventilated; or

.2 the spaces are maintained in an inert condition.

Susceptibility to leakage (see 4.4)

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.1.2 the spaces are adequately ventilated; or

.2 the spaces are maintained in an inert condition.

Susceptibility to leakage (see 4.4)

A-23 A risk assessment should be conducted to ensure that risks arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship are addressed. Consideration should be given to the hazards associated with properties of liquefied hydrogen and hydrogen gas, physical layout, operation and maintenance, following any reasonably foreseeable failure. For the risk assessment, appropriate methods, e.g. HAZID, HAZOP, FMEA/FMECA, what-if analysis, etc., should be adopted taking into account IEC/ISO 31010:2019 "Risk management Risk assessment techniques"7) and SAE ARP 5580 -2001 "Recommended failure modes and effects analysis (FMEA) practices for non-automobile applications"9).

General (see 4.1)

A-24 Relief valve sizing should be undertaken for the most onerous

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applications"9).
General (see 4.1)

A-24 Relief valve sizing should be undertaken for the most onerous scenario. The evaluation should include the fire scenario and should consider the resulting magnitude of the heat flux on the cargo containment system.

High pressure hazard

(see 4.8)
A-25 A filling limit exceeding 98% at reference temperature should not be permitted.
High pressure hazard (see 4.8)
A-26 Bolted flange connections of hydrogen piping should be avoided where welded connections are feasible.
Susceptibility to leakage (see 4.4.2)
A-27 Due consideration should be given to the invisible nature of hydrogen fire from the viewpoint of safety of ships and especially personnel in case of fire.
Fire hazard (see 4.7.1)

3 EXPLANATION ON GENERAL REQUIREMENTS

3.1 Properties of liquefied hydrogen

The application of general requirements in the Code for liquefied hydrogen has been

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3.1 Properties of liquefied hydrogen

The application of general requirements in the Code for liquefied hydrogen has been
considered based on a comparison study on the physical properties of liquefied hydrogen and LNG. LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate flammable high pressure gas. For reference, table 3 shows the comparison of physical properties of hydrogen and methane, the major component of LNG.

Table 3: Comparison of physical properties of Hydrogen and Methane

	Hydrogen	Methane	References
Boiling temperature (K)*	20.3	111.6	ISO1), Annex A, Table A.3
Liquid density (kg/m3)*	70.8	422.5	ISO1), Annex A, Table A.3
Gas density (kg/m3)**	(Air: 1.198) 0.084	0.668	NIST RefProp10)
Viscosity (g/cms x 10-6)			
Gas	8.8	13.49	
Liquid	10.91	116.79	

NIST RefProp10)
NIST RefProp10)

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URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
Viscosity (g/cms x 10-6)
Gas
Liquid

8.8
13.49

10.91
116.79

NIST RefProp10)
NIST RefProp10)
Flame temperature in air (°C) 2396 2230 Calculated using Cantera and GRI 3.0 mechanism
Maximum burning velocity (m/s) 3.15 0.385 Calculated using Cantera and GRI 3.0 mechanism
Heat of vaporization (J/g)* 454.6 510.4 ISO1), Annex A, Table A.3
Lower flammability limit
(% vol. fraction)*** 4 5.3 ISO1), Annex B, Table B.2
Upper flammability limit
(% vol. fraction)*** 77 17.0 ISO1), Annex B, Table B.2
Minimum ignition energy (mJ)*** 0.017 0.274 ISO1), Annex B, Table B.2
Auto-ignition temperature*** 858 810 ISO1), Annex B, Table B.2
Toxicity Non Non Orange book5)
Temperature at critical point (K) 33.19**** 190.55
Hydrogen: ISO1), Annex A,
Table A.1
Methane: The Japan
Society of Mechanical
Engineers, Data Book,
Thermophysical
Properties of Fluids (1983)
Pressure at critical point (kPaA) 1315**** 4595
Hydrogen: ISO1), Annex A,
Table A.1
Methane: The Japan

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URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.565(108).pdf
Properties of Fluids (1983)
Pressure at critical point (kPaA) 1315**** 4595
Hydrogen: ISO1), Annex A,
Table A.1
Methane: The Japan
Society of Mechanical
Engineers, Data Book,
Thermophysical

Properties of Fluids (1983)

Remarks: * At their normal boiling points for comparison purpose.
** At normal temperature and pressure.
*** Ignition and combustion properties for air mixtures at 25°C
and 101.3 kPaA.
**** Normal Hydrogen.

3.2 Explanation on respective requirements

3.2.1 Ship type (column 'c')

As the hazard associated with hydrogen cargo is flammability but not toxicity, the ship type is considered 2G.

3.2.2 Independent tank type C required (column 'd')

Independent tank type C is allocated only to dangerous goods of class 2.3 whose vapour density is heavier than air. Independent tank type C is considered not to be required for liquefied hydrogen.

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MSC 108/20/Add.1
Annex 20, page 9

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